

2. multinomial theorem

$$(x_1 + x_2 + \dots + x_r)^n$$

$$= \sum \binom{n}{n_1, n_2, \dots, n_r} x_1^{n_1} x_2^{n_2} \dots x_r^{n_r}$$

$$n_1, n_2, \dots, n_r: n_1 + n_2 + \dots + n_r = n$$

proof:

$$\textcircled{1} (x_1^{(1)} + x_2^{(1)} + \dots + x_r^{(1)}) (x_1^{(2)} + x_2^{(2)} + \dots + x_r^{(2)}) \dots (x_1^{(n)} + x_2^{(n)} + \dots + x_r^{(n)})$$

by induction, (1) is proved. $\square$

r^n terms.

1.4 (method 2) (combinatorial proof)

$$\textcircled{2} n_1 \text{ of } x_1^{(1)}\text{'s, } n_2 \text{ of } x_2^{(1)}\text{'s,}$$

$$\textcircled{1} (x_1^{(1)} + x_2^{(1)}) (x_1^{(2)} + x_2^{(2)}) \dots (x_1^{(n)} + x_2^{(n)})$$

$$n_r \text{ of } x_r^{(1)}\text{'s : } \binom{n}{n_1, n_2, \dots, n_r} \text{ terms.}$$

2^n terms,

$$\textcircled{3} \text{ let } x_1^{(1)} = x_1^{(2)} = x_1^{(n)} = x_1$$

$$x_2^{(1)} = x_2^{(2)} = x_2^{(n)} = x_2$$

$$x_r^{(1)} = x_r^{(2)} = x_r^{(n)} = x_r$$

$$\therefore (x_1 + x_2 + \dots + x_r)^n = \sum \binom{n}{n_1, n_2, \dots, n_r} x_1^{n_1} x_2^{n_2} \dots x_r^{n_r}$$

$$n_1, n_2, \dots, n_r: n_1 + n_2 + \dots + n_r = n$$

$\square$

$$= \sum_{i=1}^{n-1} \binom{n-1}{i-1} + \binom{n-1}{i} x^i y^{n-i}$$

$$+ y^n + x^n$$

$$= \sum_{i=0}^n \binom{n}{i} x^i y^{n-i}$$

$$(\text{here } \binom{n-1}{i-1} + \binom{n-1}{i} = \binom{n}{i})$$

Pascal's identity)

by induction, (1) is proved. $\square$

binomial theorem multinomial theorem

1. binomial theorem

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k} \quad (1)$$

1.3 proof: (method 1), (by induction)

$$\textcircled{1} \text{ if } n=1, (1) \text{ is } x+y = \binom{1}{0} x^0 y^1 + \binom{1}{1} x^1 y^0 = x+y$$

$\textcircled{2}$ assume (1) for $n-1$,

$$\textcircled{3} (x+y)^n = (x+y)(x+y)^{n-1}$$

$$= (x+y) \sum_{k=0}^{n-1} \binom{n-1}{k} x^k y^{n-1-k}$$

$$= \sum_{k=0}^{n-1} \binom{n-1}{k} x^{k+1} y^{n-1-k} + \sum_{k=0}^{n-1} \binom{n-1}{k} x^k y^{n-k}$$

$$= \sum_{i=1}^n \binom{n-1}{i-1} x^i y^{n-i} + \sum_{i=0}^n \binom{n-1}{i} x^i y^{n-i}$$

$\square$